APPLICATIONS

malware clustering w/ sdhash

in the beginning …

4.15 million files …

1.6 TB …

All named ‘malware.zip’ …

Submitted by random people …

How do we make sense of them?

Actually, how do we triage them?

idea

We have this ~~hammer~~ sdhash

- » let's find some nails!

First, some housekeeping & benchmarking

- » unzip (30+ hrs, 48c)
- » sdhash (13 hrs, 48c)
- » 9000 directories

  1000-4294 —> ~1,000 files/512MB

  sdhash-gpu: 8s (gtx680)

  4295-9999 —> ~100/<100MB

  sdhash-gpu: 0.5s

rationale

If A is bad && A ~ B, then B ~ bad

Things to do

» clustering

grab a sample and test for self-similarity

» labelling

find known samples (by md5)

propagate label by similarity

trawl #1

9999 → 108 files → 56MB

Run *virustotal* → find labeled samples

sdhash & compare vs. entire set

- » 65m on gtx680
  - 115k sample comps/sec
  - [18m min for any query; bring more data!]
- » 96/108 matched something
- » estimated 400k significant results

#1 breakdown

```
128047 9999-8109-3  W32/Yuner.A/W32.Badday
128047 9999-7712-8  W32/Yuner.A/W32.Badday
128047 9999-6560-6  W32/Yuner.A/W32.Badday
128047 9999-3222-3  W32/Yuner.A/W32.Badday
119273 9999-7671-9  Cosmu/Overdoom/Lamer
 47552 9999-6602-8  Worm:Win32/Allaple.A
 47552 9999-3748-2  Worm:Win32/Allaple.A
 47550 9999-5125-4  Worm:Win32/Allaple.A
 41743 9999-2604-6  Backdoor:Win32/Small (Udr)
 33181 9999-0217-3  Virus:Win32/Krepper.30760
…
  2723 9999-3106-7  Worm:Win32/Kucirc.A (Cosmu?)
  2720 9999-1233-9  TrojanDownloader:Win32/VB.PX
  2441 9999-0522-5  Virus:Win32/Virut.AK (Allaple?)
  2190 9999-5943-7  Worm:Win32/Vobfus.gen!D
  2170 9999-9278-2  Worm:Win32/Vobfus.gen!D
  2083 9999-5743-2  Backdoor.Poison (Ivy?)
  2075 9999-2737-0  Trojan:Win32/Agent
…
```

#2: zeus

Grab zeustracker binaries (1.6G)

sdhash & comp → internal similarity

Take 50 random samples, trawl

- 26m (gpu), 1157 matches, 932 **not** in tracker

Go for the full sample

- 28h30m (2 x gpu)
- conf >=75 —> 15,965 (4926 trackers)
- conf >=40 —> extra 8967 (4546)

#3: other celebrities

Get known md5s from web

» duqu, stuxnet, rustock

Find md5 match in data set

sdhash compare vs. whole set

» 40min, gtx680

```
# components of duqu match each other:
4050-7069-81   1774-9632-72   79
8385-0091-9    2915-4761-7    59

# found a -different- duqu dropper
1774-9632-72   1247-3173-60   45

# duqu vs stuxnet.B/C
4050-7069-81   3724-2013-53   26
```

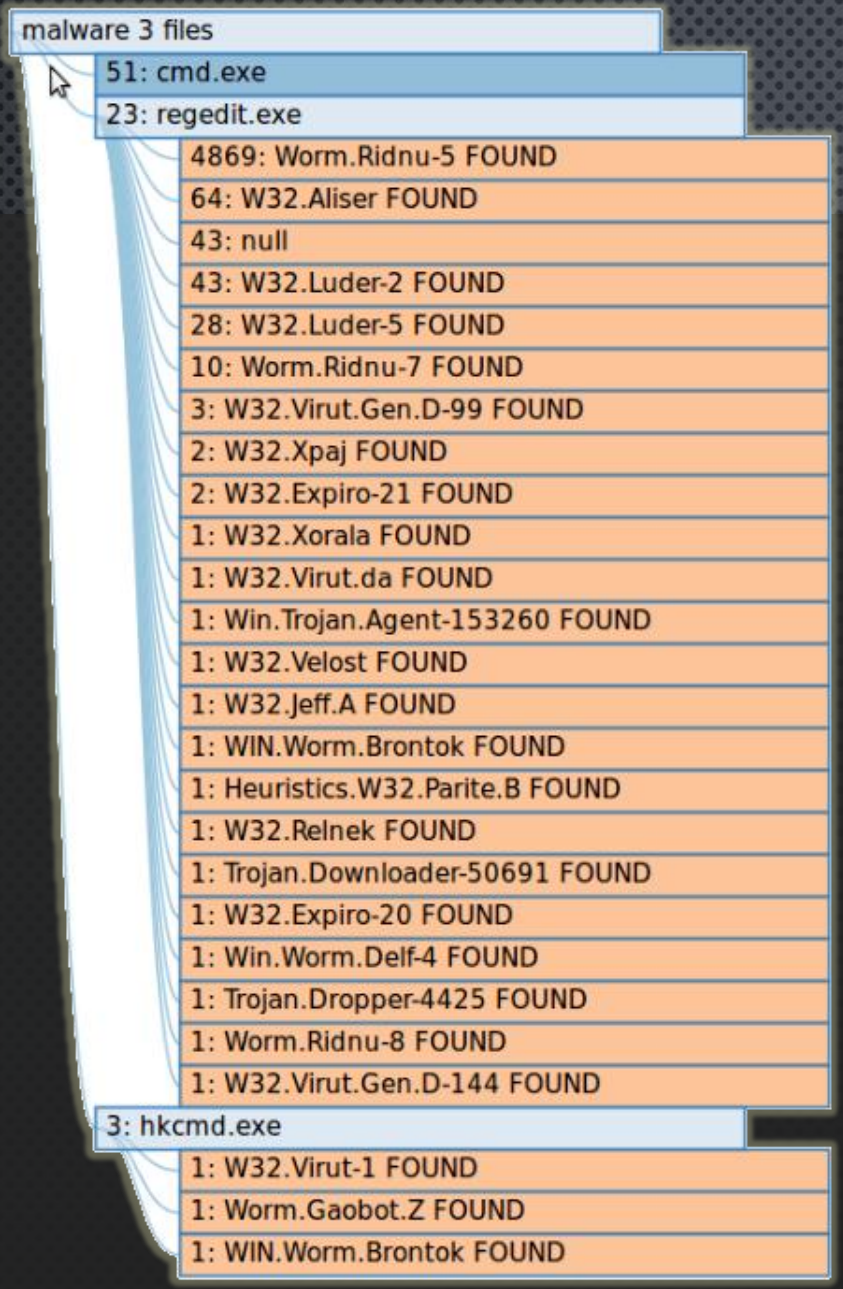
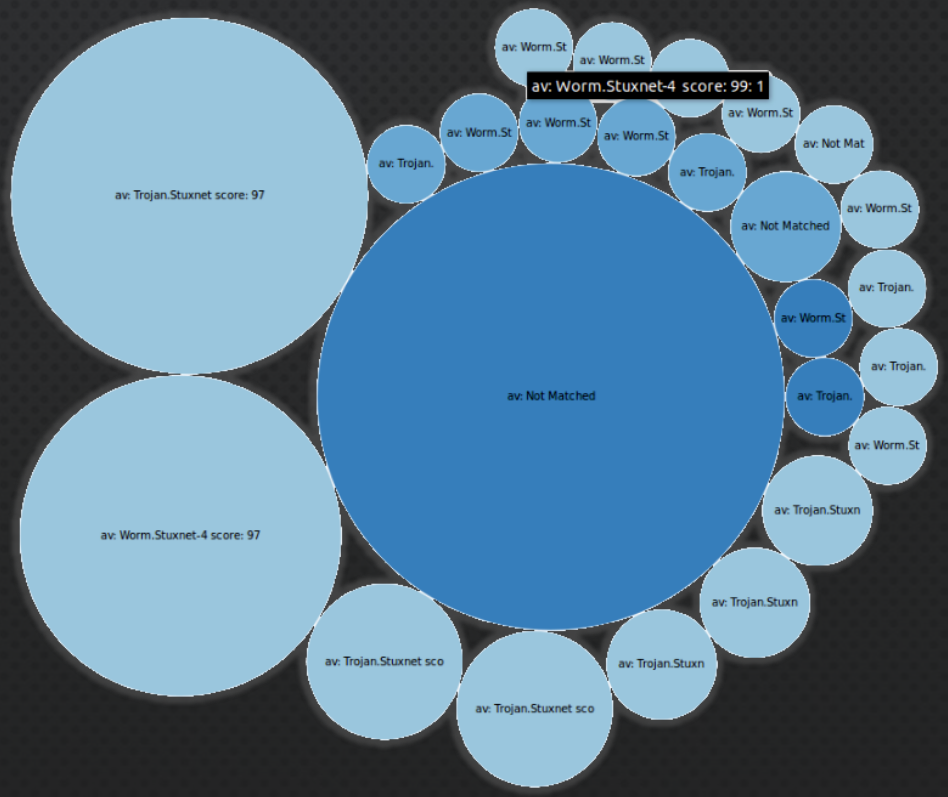
#3 breakdown

```
>= 40

   1477 1661-5901-80   Rustock dropper
     37 3454-7622-37   Stuxnet.A or B
     60 3567-1746-42   Stuxnet.A (dropper)
      5 4161-3954-81   Stuxnet.A
      3 4050-7069-81   Duqu
      3 1774-9632-72   Duqu
      2 8385-0091-9    Duqu
      2 2915-4761-7    Duqu
```

#4: windows is malware?

```
5650 ./WINDOWS/regedit.exe
2848 ./WINDOWS/system32/cmd.exe
2719 ./WINDOWS/system32/regsvr32.exe
2408 ./WINDOWS/system32/rundll32.exe
1912 ./WINDOWS/system32/reg.exe
1724 ./WINDOWS/system32/notepad.exe
1473 ./WINDOWS/system32/dwwin.exe
1260 ./WINDOWS/ie8/iexplore.exe
1169 ./WINDOWS/system32/logonui.exe
1069 ./WINDOWS/system32/wextract.exe
1033 ./WINDOWS/explorer.exe
 841 ./WINDOWS/system32/net1.exe
 648 ./WINDOWS/system32/netsh.exe
 575 ./WINDOWS/Help/Tours/mmTour/tour.exe
 566 ./WINDOWS/system32/net.exe
 529 ./WINDOWS/system32/at.exe
 522 ./WINDOWS/system32/migpwd.exe
 516 ./WINDOWS/system32/ping.exe
 …
```

63,352 matches
378 unique

zombie Windows executables